Using deep learning to generate synthetic tax records

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The Budget Lab at Yale

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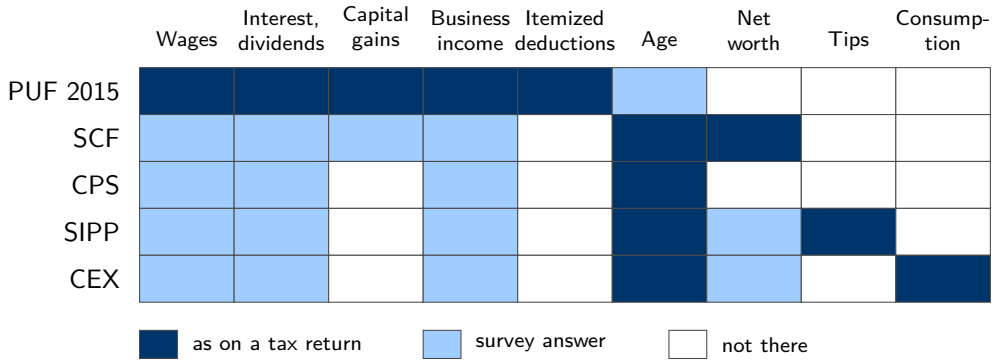
What Tax-Simulator needs

Married	Dependents	Age	Wages	Interest	Cap. gains	Net worth	Weight
yes	2	44	$142,000	$310	$5,200	$610,000	812
no	0	27	$38,000	$0	$0	$9,000	1,404
yes	0	71	$0	$4,900	$61,000	$2,400,000	655
... about 200 million more							

- Realistic records: the variables go together the way they do in life
- The right totals for the year
- Every year, 2015 to 2060

Made-up records.

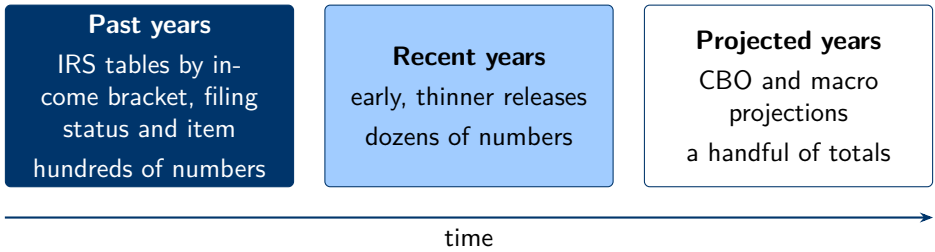
The information is scattered



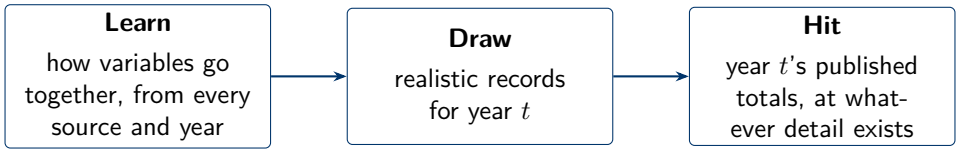
No source has everything. Each covers different years.

Simplified.

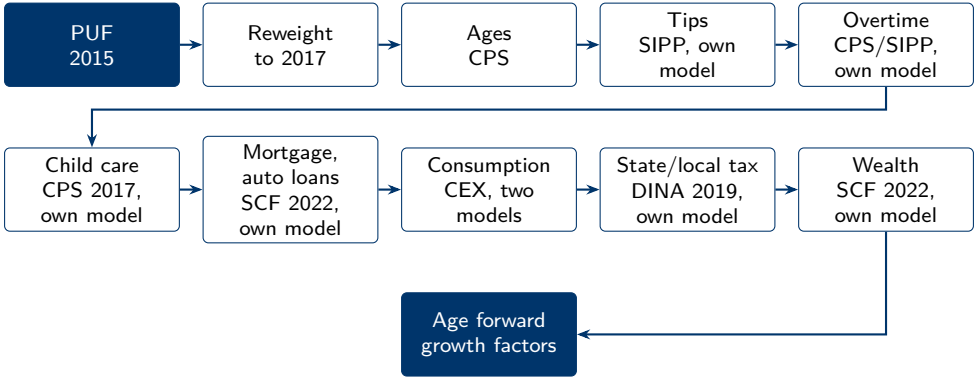
Targets come at different levels of detail



The task


Learn
how variables go together, from every source and year
Draw
realistic records for year t
Hit
year t's published totals, at whatever detail exists

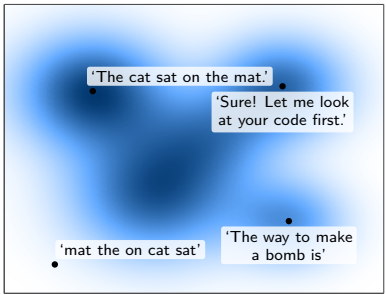
What we did before



- Each step sees only a few variables
- Variables come from different years: 2015, 2017, 2019, 2022
- Each period is aged by its own method

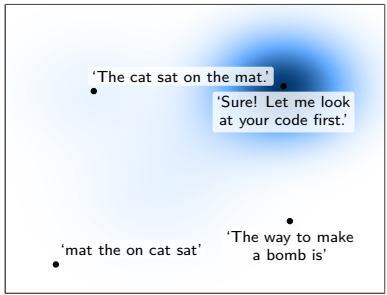
Language models: pretraining, then post-training

Pretraining



Measure how language is distributed: which sentences are likely

Post-training

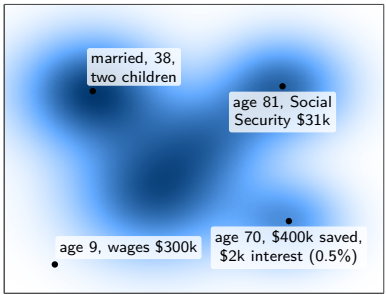


Squeeze the mass toward the answers we want

Each point is a sentence; darker means more likely. Illustration.

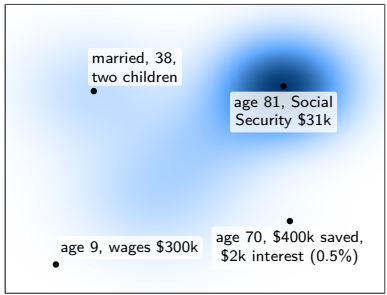
Our model: pretraining, then post-training for 2040

Pretraining



Measure how tax units are distributed: which records are likely

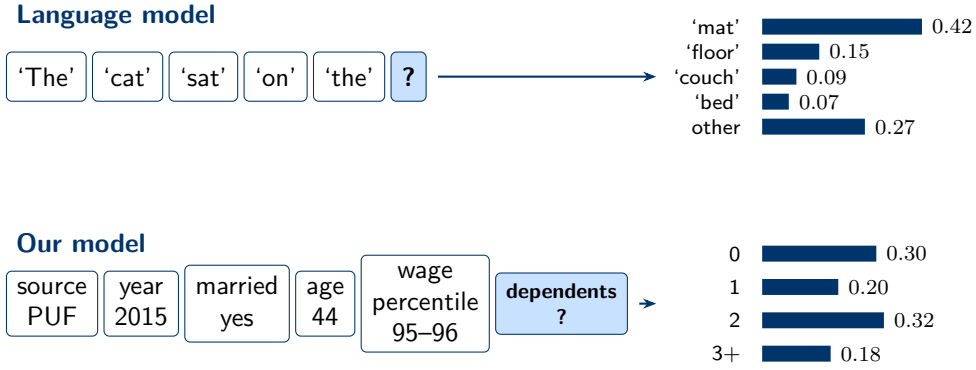
Post-training for 2040



Shift the mass toward 2040: more older units, fewer retirees on 2010s interest rates

Each point is a tax unit; darker means more likely. Illustration.

Predicting the next word, predicting the next variable



Made-up probabilities.

Training, step by step

Training data: real records from the PUF and the SCF

source	year	married	dependents	wages	interest	dividends	net worth	UI benefits
PUF	2015	yes	2	$142k	$310	$2.4k	–	$0
PUF	2015	no	0	$38k	$0	$0	–	$4.2k
SCF	2019	no	1	$48k	$120	$0	$85k	–
SCF	2016	yes	2	$95k	$350	$1.1k	$610k	–



Made-up records and numbers.

Training, step by step

Hide a random set of cells (source and year always stay visible)

source	year	married	dependents	wages	interest	dividends	net worth	UI benefits
PUF	2015	yes	?	$142k	?	$2.4k	–	$0
PUF	2015	?	0	?	$0	$0	–	?
SCF	2019	no	1	$48k	?	$0	?	–
SCF	2016	yes	2	?	$350	$1.1k	$610k	–



Made-up records and numbers.

Training, step by step

The network guesses a distribution for each hidden cell

source	year	married	dependents	wages	interest	dividends	net worth	UI benefits
PUF	2015	yes		$142k		$2.4k	–	$0
PUF	2015		0		$0	$0	–	
SCF	2019	no	1	$48k		$0		–
SCF	2016	yes	2		$350	$1.1k	$610k	–

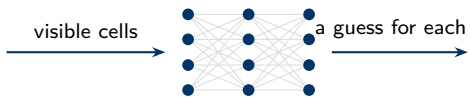


Made-up records and numbers.

Training, step by step

Compare with the truth (dark bar)

source	year	married	dependents	wages	interest	dividends	net worth	UI benefits
PUF	2015	yes	2	$142k	$310	$2.4k	–	$0
PUF	2015	no	0	$38k	$0	$0	–	$4.2k
SCF	2019	no	1	$48k	$120	$0	$85k	–
SCF	2016	yes	2	$95k	$350	$1.1k	$610k	–



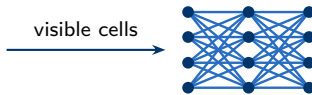
probability on the truth: 0.21

Made-up records and numbers.

Training, step by step

Nudge every weight so the truth gets more probability

source	year	married	dependents	wages	interest	dividends	net worth	UI benefits
PUF	2015	yes	2	$142k	$310	$2.4k	–	$0
PUF	2015	no	0	$38k	$0	$0	–	$4.2k
SCF	2019	no	1	$48k	$120	$0	$85k	–
SCF	2016	yes	2	$95k	$350	$1.1k	$610k	–



update 1
probability on the truth: 0.21

Made-up records and numbers.

Training, step by step

Next batch: hide a different random set

source	year	married	dependents	wages	interest	dividends	net worth	UI benefits
PUF	2015	yes	2	?	$310	?	–	?
PUF	2015	no	?	$38k	$0	$0	–	$4.2k
SCF	2019	?	1	?	$120	$0	$85k	–
SCF	2016	yes	2	$95k	?	$1.1k	?	–

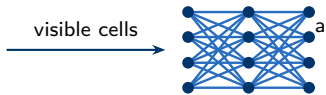


Made-up records and numbers.

Training, step by step

Guess, compare, nudge again

source	year	married	dependents	wages	interest	dividends	net worth	UI benefits
PUF	2015	yes	2	$142k	$310	$2.4k	–	$0
PUF	2015	no	0	$38k	$0	$0	–	$4.2k
SCF	2019	no	1	$48k	$120	$0	$85k	–
SCF	2016	yes	2	$95k	$350	$1.1k	$610k	–



update 2
probability on the truth: 0.27

Made-up records and numbers.

Training, step by step

After about 10,000 updates, 4,096 records each

source	year	married	dependents	wages	interest	dividends	net worth	UI benefits
PUF	2015	yes	2	$142k	$310	$2.4k	–	$0
PUF	2015	no	0	$38k	$0	$0	–	$4.2k
SCF	2019	no	1	$48k	$120	$0	$85k	–
SCF	2016	yes	2	$95k	$350	$1.1k	$610k	–



Made-up records and numbers.

Training: fill in the blanks

- Hide, guess, compare, nudge: about 10,000 times
- The blanks can be anywhere, so one model answers every question of the form "this variable, given those"
- A variable a source never has (grey: net worth on the PUF, UI benefits on the SCF) is never asked

Writing a record, one variable at a time

A sentence	A record
The	PUF, 2015
The cat	PUF, 2015, married: yes
The cat sat	PUF, 2015, married: yes, wages: $142k
The cat sat on the mat	. . . , deps: 2, interest: $310, net worth: $610k

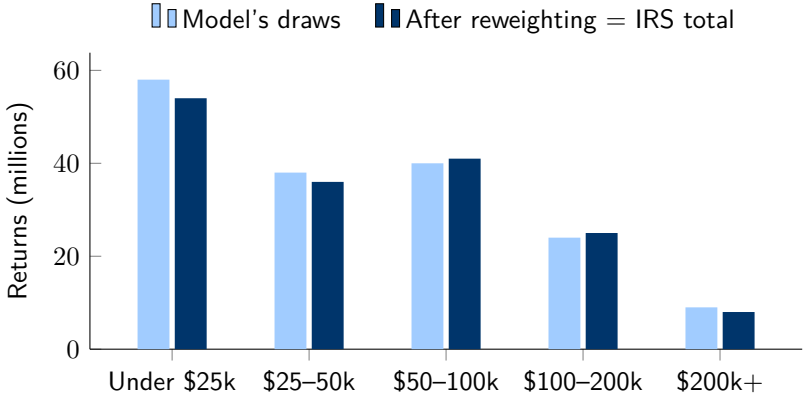
Each step: ask the model for the probabilities, draw one, add it to the context.

Made-up records.

Why deep learning

- **One model, every question.** No chain of separate imputations.
- **It finds the interactions itself.** Nonlinear, many variables at once.
- **Gaps are just blanks.** Every source trains the same model, whatever it is missing.
- **Source and year are part of the prompt.** Ask for a record "as the 2015 PUF would show it" or "as the 2019 SCF would".

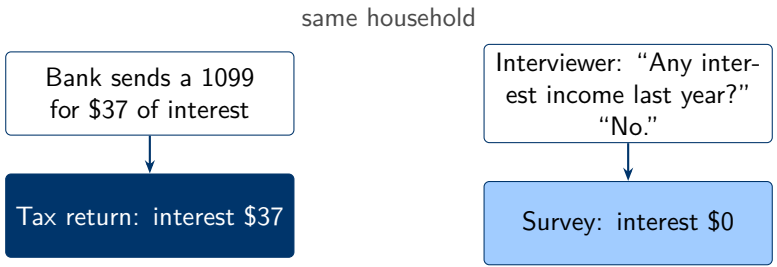
Post-training: hit the totals



Change the weights as little as possible while matching every total.

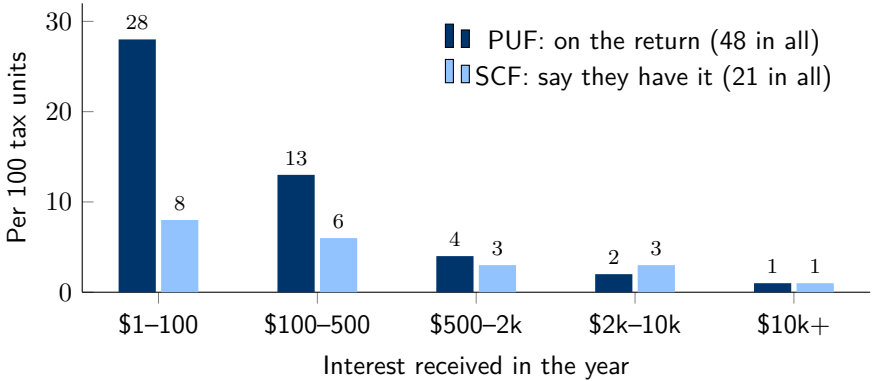
Made-up numbers.

Challenge: the same word means different things



Population	filers or everyone; tax units or households
Definition	what counts: e.g. losses capped at $3,000 on a return
Measurement	small amounts forgotten, misremembered, rounded

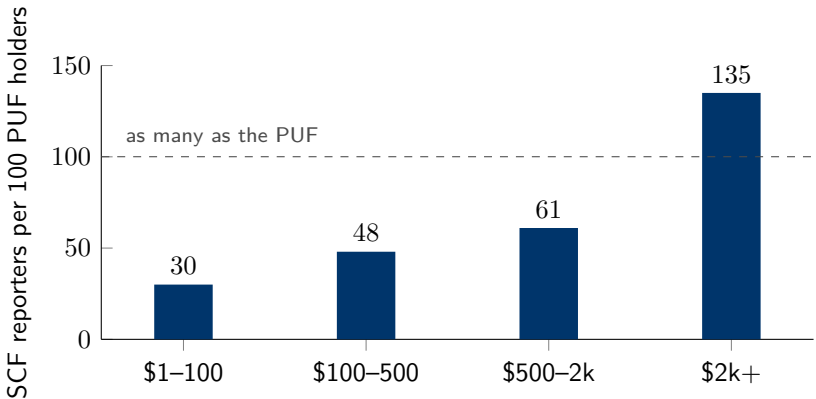
Interest: how many hold any?



The gap is all in small amounts. Above $2,000 the two sources agree.

Married, wages $75–200k. 2015 PUF and 2016 SCF (tax year 2015), weighted.

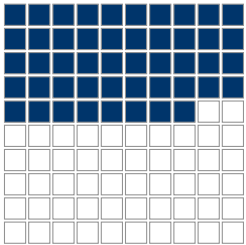
Small amounts go unreported



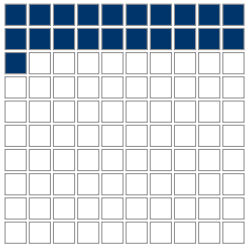
Of every 100 couples whose return shows \$1–100 of interest, only about 30 tell the SCF they have any.

Married, wages \$75–200k. 2015 PUF, 2016 SCF, weighted. Above 100: no fill-in.

Why it matters: what interest says about wealth



PUF: 48 of 100 hold interest



SCF: 21 of 100 say they do

- Wealth comes only from the SCF: the model learns what interest holders own from the 21
- It then applies that to the PUF's 48. But the 21 lean toward larger amounts
- In our first test, taking SCF answers at face value put net worth off by a factor of two

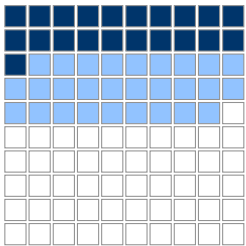
Married, wages $75–200k. 2015 PUF and 2016 SCF, weighted.

The fix, step 1: two slots

SCF household	interest, as said	interest, as on a return
household 1	$250	$250
household 2	$0	?
household 3	$0	?
household 4	$1,800	$1,800

- The answer the household gave is kept, never changed
- A second slot holds what its tax return would show
- Reporters copy their answer. Zero-reporters get filled in, some with a small amount
- PUF records have only the second slot, so both sources now mean the same thing by "interest"

The fix, step 2: how many to fill in



SCF, after the fill-in

reported filled in none

Per 100 couples	PUF	SCF	Fill in
$1–100	28	8	20
$100–500	13	6	7
$500–2k	4	3	2
$2k and over	3	3	0
Any	48	21	28

The PUF says how many hidden holders there are, and at what sizes. Of the 79 couples who said none, about 28 get a small amount.

Married, wages $75–200k. 2015 PUF and 2016 SCF, weighted. Rows may not add because of rounding.

The fix, step 3: which households

For \$1–100: 30 report for every 100 holders, so each reporter stands for $100/30 - 1 \approx 2.3$ hidden holders.

Couple A: said no interest

\$300k in the bank, owns stocks

Model: looks like a reporter of \$1–100 with chance 20%, like a zero with 60%

Chance of a hidden holding:
$20\% \times 2.3 \div 60\% \approx$ **78%**

Couple B: said no interest

\$3k net worth, renter

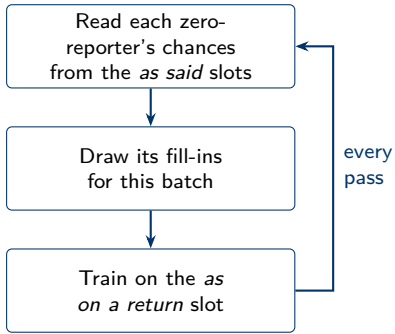
Model: looks like a reporter of \$1–100 with chance 2%, like a zero with 95%

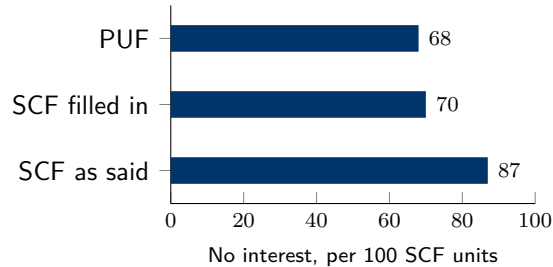
Chance of a hidden holding:
$2\% \times 2.3 \div 95\% \approx$ **5%**

Households that look like reporters get most of the fill-ins. The PUF sets how many; the rest of the SCF record sets who.

Couples A and B are made up; the 30 per 100 is real.

The fix, step 4: inside training





The fill-ins never feed back into the chances, so the model does not learn from its own guesses.

2016 SCF units (tax year 2015); PUF predicted at the same household types. Latest model, 21 October.

What the fix assumes, and how well it works

The assumption

Whether a holder reports depends only on its type (married, dependents, wages) and how much it holds. Nothing else on the record.

A test with a known answer

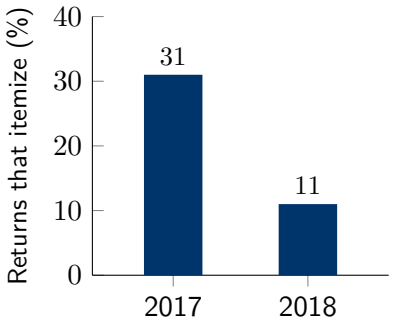
Make PUF records answer the way SCF households do, then try to undo it.

- True share with no interest: 71.6%. Recovered: 70.7–73.0%
- If reporting also depends on something the record does not show, it overcorrects by about 8%

On the real SCF

Still about 1.7 points short of the PUF for interest: some households come out with a chance above 100%, and are capped

Challenge: the return itself changes



- The 2017 tax law cut itemizing by two thirds
- A 2015 return shows mortgage interest for people who no longer report it
- New items, like the tips and overtime deductions, are on no past return

Approximate shares.

Other challenges

Different populations

The PUF has no non-filers. Surveys count households, not tax units.

A thin top

Few very rich records in any source.

Future years

No microdata. Only targets, and a choice of which past years to lean on.

Checking the result

No source has every variable, so some relationships have no truth to compare against.

Where we are

- A 2019 file: every PUF variable plus a 20-item balance sheet
- Trained on the 2015 PUF and four SCF waves (2013–2022)
- Hits about 285 targets: IRS by income bracket, IRS business income, SCF wealth
- Median net worth $90,900 against the SCF's $91,050

What is left

- More surveys: CPS, CEX, SIPP
- Projected years
- A tax calculator inside post-training, for targets on computed tax
- A standing set of checks
- Replace the current pipeline as Tax-Simulator's input

Questions